

Realistic Clothed Human and Object Joint Reconstruction from a Single Image

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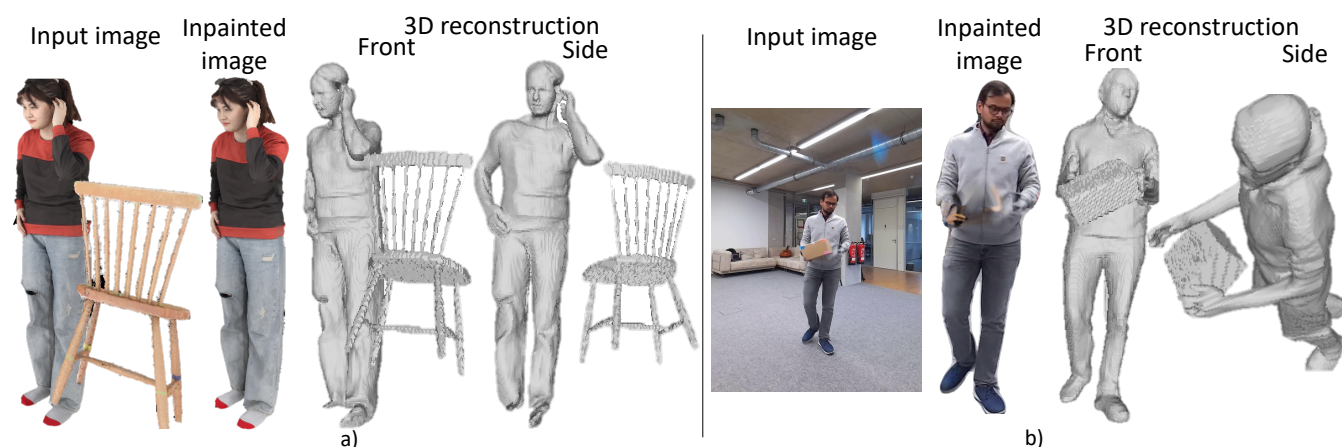


Figure 1: ReCHOR jointly reconstructs realistic clothed humans and objects from (a) synthetic and (b) real images.

Abstract

In recent years, there has been a growing interest towards developing personalized human avatars, for applications ranging from virtual reality, movie production, gaming and social telepresence. In the future, these avatars will be expected to also interact with everyday objects. Achieving this requires not only accurate human reconstruction, but also a joint understanding of surrounding objects and their interactions, making 3D human-object reconstruction key to its success. Existing methods that can jointly reconstruct 3D humans and objects from a single RGB image produce only coarse or template-based shapes, thus failing to capture realistic details in the reconstruction, such as loose clothing on the human body. In this work, for the first time, we propose an approach to jointly

reconstruct 3D clothed humans and objects, given a monocular image of a human-object scene.

At the core of our framework, is a novel attention-based model that jointly learns an implicit function for the human and the object. Given a query point, our model utilizes pixel-aligned features from the input human-object image as well as from separate, non-occluded views of the human and the object as synthesized by a diffusion model. This allows the model to reason about human-object spatial relationships as well as to recover details from both visible and occluded regions, enabling realistic reconstruction. To guide the reconstruction, we condition the neural implicit model on human-object pose estimation priors. To support training and evaluation, we also introduce a synthetic human-object dataset. We demonstrate on real-world datasets that our approach significantly improves the perceptual quality of 3D human-object reconstruction.

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CVMP '25, London, United Kingdom

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ACM ISBN 979-8-4007-2117-5/25/12
<https://doi.org/10.1145/3756863.3769705>

CCS Concepts

• Computing methodologies → Reconstruction.

Keywords

3D Human-Object Reconstruction, 3D Human-Object Interaction, Non-parametric Reconstruction, Implicit Reconstruction

ACM Reference Format:

Ayushi Dutta, Marco Pesavento, Marco Volino, Adrian Hilton, and Armin Mustafa. 2025. Realistic Clothed Human and Object Joint Reconstruction from a Single Image. In *European Conference on Visual Media Production (CVMP '25)*, December 03–04, 2025, London, United Kingdom. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3756863.3769705>

1 Introduction

Recently, there has been a continued interest in developing personalized digital avatars, driven by applications in virtual reality, film production, gaming, and social telepresence. These avatars are expected to faithfully represent real humans, not only in their appearance but also in their ability to interact naturally with objects. Looking ahead, such interactions will form the basis of immersive experiences, where users can seamlessly engage with digital content as with the physical world. This vision places 3D human–object reconstruction at the center of research in computer vision and graphics.

Existing methods [Nam et al. 2024; Xie et al. 2024a, 2022, 2023; Zhang et al. 2020] for human-object reconstruction have demonstrated that when given an image depicting both humans and objects, their joint 3D reconstruction accurately reflects their mutual spatial relationship and interaction. However, such methods produce only coarse or template-based shapes, neglecting the free-form geometry of human-object shapes such as clothing details on the human, thereby reducing the realism of the reconstructed scene. Motivated by this, we aim to jointly reconstruct realistic clothed humans and objects from a single-view human-object scene.

However, this task is inherently ill-posed, presenting significant challenges. First, the human-object spatial configuration such as 3D location, shape, and relative pose need to be inferred which is difficult when only a single image is available. Multiple 3D spatial configurations of humans and objects can project to the same 2D image, leading to depth-scale ambiguity and human-object occlusion further complicates the inference. Second, when aiming for realism, the reconstruction must account for surface-level details, which are particularly challenging to estimate under occlusion. This difficulty is compounded by the lack of training data, as realistic reconstruction relies on high-quality ground-truth meshes that are hard to acquire at scale.

We thereby propose ReCHOR, a novel framework for Realistic Clothed Human and Object joint Reconstruction from a single image. Our core idea is to learn an implicit function for the human and the object jointly, allowing unconstrained surface geometry and enabling realistic reconstruction while considering their spatial relationship. To aid this learning, we propose an attention-based neural implicit model that attends to pixel-aligned features from the input image for understanding human-object spatial relationship, as well as to individual, non-occluded views of the human and the object as generated by a diffusion model, to facilitate the retrieval of fine details particularly in occluded regions.

However, implicit representations, which depend on strong feature evidence and do not have any geometric constraints, can fail to generalize well to unknown poses and result in unnatural shapes and poses in their reconstructions. Following prior works [Ho et al. 2024; Xiu et al. 2023] that incorporate 3D spatial regularization, we

condition the neural implicit model on human-object pose estimation priors obtained from coarse or template-based human-object reconstruction methods, encouraging plausible reconstructions. Moreover, query points for implicit function learning need to be sampled within a fixed 3D volume. Most works on implicit function learning assume a weak perspective camera model, which is not suitable for human-object reconstruction. We thus learn the representation in the reference frame of the pose prior, using a perspective projection model.

Finally, to tackle the challenge of the lack of availability of scalable datasets with high-quality 3D ground-truth meshes for human-object scenes, we introduce synHOR, a simple, yet effective, synthetic dataset to train and evaluate ReCHOR. Specifically, we generate several 3D spatial configurations of human-object scenes by randomly placing 3D human scans from THuman2.0 [Yu et al. 2021] with selected object meshes from BEHAVE [Bhatnagar et al. 2022] and HODome [Zhang et al. 2023a].

Our key technical contributions are:

- **ReCHOR Framework:** We present the first novel approach to address the challenging task of jointly reconstructing realistic clothed humans and objects from single images.
- **Attention-based neural implicit model:** We estimate the joint implicit representation of clothed humans and objects. Pixel-aligned features are extracted from different image contexts along with 3D spatial priors via transformer encoders, capturing realistic details while learning clothed human-object spatial relationship.
- **Clothed human-object dataset:** We contribute a scalable synthetic dataset to facilitate training and evaluation of realistic clothed human-object reconstructions.
- **Evaluation:** We also evaluate our model on the real world dataset, BEHAVE [Bhatnagar et al. 2022] and demonstrate superior reconstruction quality compared to state-of-the-art methods. We thus set the first benchmark for realistic clothed human-object reconstruction from single images.

2 Related works

2.1 3D human and object reconstruction

To reconstruct 3D human and object jointly, previous methods use parametric models to fit human and object meshes satisfying various constraints. PHOSA [Zhang et al. 2020] and D3D-HOI [Xu et al. 2021] each proposed an optimization based framework with physical constraints on scale and predefined contact priors. Wang et al. [Wang et al. 2022] modeled 3D human-object shapes from an image using commonsense knowledge from large language models. Holistic++ [Chen et al. 2019] modeled fine-grained human-object relations in a scene using Markov chain Monte Carlo method. CHORE proposed to fit a parametric model to a learned neural-implicit functions. Vistracker [Xie et al. 2023], InterTrack [Xie et al. 2024b] reconstruct human-object from a single video, by specifically modeling the temporal context. Recently, CONTHO [Nam et al. 2024] proposed a method to refine human-object reconstruction from an image by 3D guided contact estimation. ProciGen [Xie et al. 2024a] proposed a Hierarchical Diffusion Model to reconstruct human and object. None of the current approaches for single-view human-object reconstruction can recover realistic clothed humans

at the same fidelity as ReCHOR. Alternate methods [Jiang et al. 2022, 2023; Sun et al. 2021; Zhang et al. 2023a] that can reconstruct clothed human-object, either use sparse, multi-views or monocular RGBD image as inputs, thereby relaxing their constraints and are thus not comparable to our work.

2.2 Neural implicit model for 3D reconstruction

Early reconstruction methods cannot produce realistic 3D shapes due to the discretized nature of traditional representations like voxels, meshes or point clouds. The introduction of the implicit representation for 3D reconstruction [Chen and Zhang 2019; Mescheder et al. 2019; Park et al. 2019] has led deep learning approaches to adopt this continuous representation because it can represent fine details on the reconstructed shapes. These works estimate the implicit representation with neural implicit models. Several works have since used neural implicit models for object reconstruction from single images. Early models estimate occupancy [Chen and Zhang 2019; Mescheder et al. 2019] or signed distance fields [Park et al. 2019; Xu et al. 2019; Zhang et al. 2021] using multi-layer perceptrons conditioned on input features. Recent works improve the object reconstruction by incorporating prior knowledge into implicit functions [Chen et al. 2023; Liu et al. 2019]; using monocular geometric cues [Yu et al. 2022]; combining explicit templates with implicit representations [Fahim et al. 2022; Wang et al. 2023] or leveraging input global and local features [Arshad and Beksi 2023; Li and Zhang 2021]. Other implicit models focus on reconstructing high-quality 3D human shapes from single images. PIFu [Saito et al. 2019] introduced the pixel-aligned implicit function to retrieve detailed human shapes. Building on this, several works improved quality using normal maps [Saito et al. 2020; Xiu et al. 2022] or super-resolution shapes [Pesavento et al. 2022]; addressed depth ambiguity using parametric models [He et al. 2020; Zheng et al. 2021]; achieved complete reconstructions with diffusion models [Ho et al. 2024; Kolotouros et al. 2024; Sengupta et al. 2024] or incorporating additional data as depth maps [Pesavento et al. 2024; Xiu et al. 2023] or unconstrained images [Pesavento et al. 2025]. These works cannot jointly reconstruct human-object shapes since they are category-specific, with human-focused methods unable to reconstruct objects and vice-versa. ReCHOR jointly reconstructs realistic clothed humans and objects from a single image.

2.3 3D human-object datasets

Recent works BEHAVE [Bhatnagar et al. 2022], INTERCAP [Huang et al. 2024], HODome [Zhang et al. 2023a] have captured real world human-object interactions, or created synthetic dataset ProCigen [Xie et al. 2024a], to allow training and evaluation of human-object reconstruction methods. However, these datasets are unsuitable for training implicit models as they provide only parametric fits. In contrast, synHOR provides high-quality clothed human meshes for training.

3 Methodology

In the proposed pipeline ReCHOR, shown in Fig. 2, the image is first segmented to separate the human from the object, and pose priors of the SMPL-H [Romero et al. 2017] model and object are estimated. To address human occlusion, regions of the human body

that are occluded by the object in the input image are inpainted using the generative power of an image-conditioned diffusion model. Implicit representations of both the human s_h and the object s_o are then estimated using a novel attention-based neural implicit model that incorporates the input RGB image I_f , the generated full-body human image I_h , the object image I_o and the estimated pose priors.

3.1 Inpainting of occluded human body regions

This work addresses cases in images where an object occludes the human. When such occlusion happens, regions of the human body are missing in the input image, leaving the neural implicit model without the information needed to estimate these regions. We leverage the generative capability of diffusion models to inpaint the occluded body regions, using the partial human image I_p along with a mask of the occluded regions M_i . Given the input image I_f , and its estimated SMPL-H [Romero et al. 2017] model, we generate a mask of the human M_s , and obtain the human-object intersection mask M_i as $M_i = M_s - M_p$, where M_p is the segmented partial human mask.

Inspired by SiTH [Ho et al. 2024], we leverage an image-condition latent diffusion model [Rombach et al. 2022] (LDM) to learn the conditional distribution of the missing human body regions. Rather than training the LDM from scratch, we adopt a fine-tuning strategy [Kumari et al. 2023; Zhang et al. 2023b] that optimizes the cross-attention layers of a pretrained diffusion U-Net [Rombach et al. 2022]. The learning is conditioned using a ControlNet [Zhang et al. 2023b], which processes M_i to guide the generation. Additionally, we condition the U-Net with features from the partial human image I_p , extracted via a pre-trained CLIP [Ramesh et al. 2022] image encoder and a VAE encoder ϵ , to ensure that the generated image I_h matches the appearance of I_p . These features, along with randomly sampled noise ϵ , are input to the LDM model ϵ_θ , conditioned by the ControlNet, which generates a latent code z within the VAE latent distribution $z = \epsilon(I_h^{gt})$, where I_h^{gt} is the ground-truth full-body human image. The output full-body image I_h is then generated by decoding with the VAE decoder $\mathcal{D}$ a latent code $\tilde{z}$ derived through iterative denoising of Gaussian noise: $I_h = \mathcal{D}(\tilde{z})$.

We define the objective function for fine-tuning as:

$$\min_{\theta} \mathbb{E}_{z \sim \epsilon(I_h^{gt}), t, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \|\epsilon - \epsilon_\theta(z_t, t, I_p, M_i)\|_2^2 \quad (1)$$

The ground-truth latent code $z_0 = \epsilon(I_h^{gt})$ is diffused over t time steps, resulting in the noisy latent z_t . The image-conditioned LDM model ϵ_θ predicts the noise ϵ added to the noisy latent z_t , based on time step $t \sim [0, 1000]$ and conditional inputs I_p and M_i .

To inpaint the occluded body regions during inference, we generate a latent $\tilde{z}_0$ by starting with Gaussian noise $z_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and using an iterative denoising process. The final full-body image I_h is obtained with the decoder $\mathcal{D}$:

$$I_h = \mathcal{D}(\tilde{z}_0) = \mathcal{D}(f_\theta(z_T, I_h, M_i)) \quad (2)$$

where f_θ indicates the iterative denoising process of ϵ_θ . Finally, a 2D normal map S_N is estimated from I_h and concatenated with it. For simplicity, we continue to refer to this concatenated result as I_h throughout the paper.

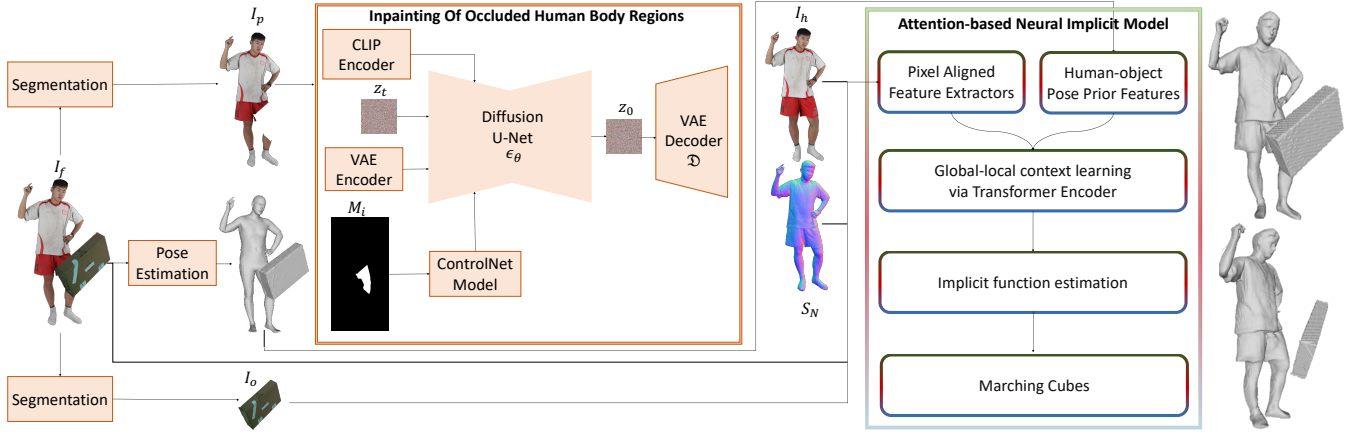


Figure 2: ReCHOR overview: Given an input image of a human-object scene, first a generative model inpaints occluded human body regions. Next, the generated image, along with an estimated normal map, the input image, the segmented object image, and estimated pose priors, are processed by an attention-based neural implicit model to jointly estimate the implicit representation of the human-object shape.

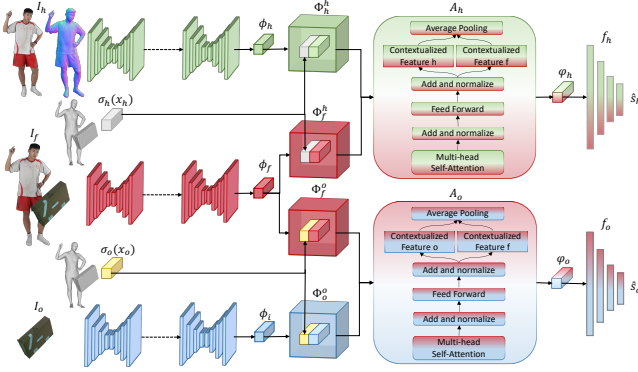


Figure 3: The attention-based neural implicit model extracts pixel-aligned features from different image contexts, merging them along with human-object pose priors through a transformer encoder to estimate the implicit representations of both humans and objects.

3.2 Attention-based neural implicit model

To reconstruct the 3D shapes of both humans and objects, we introduce an attention-based neural implicit model that jointly estimates their implicit representations. Implicit representations define a surface as the level set of a function f , i.e. $f(X) = 0$ where X is a set of 3D points in $\mathbb{R}^3$. The reconstructed surface is then defined as the zero level-set of f :

$$f' = \{x : f(x) = 0, x \in \mathbb{R}^3\} \quad (3)$$

The proposed implicit model comprises of multiple modules, as shown in Fig. 3.

3.2.1 Pixel-aligned feature extractors: Previous methods on 3D human reconstruction [Pesavento et al. 2022; Saito et al. 2020] demonstrate that projecting a 3D point $x \in \mathbb{R}^3$ in the embedded image feature space $\phi(I)$ extracted with a convolutional stacked

hourglass network significantly increases the quality of 3D human shapes. ReCHOR first extracts a feature embedding for each input image $\phi_{\{h,o,f\}}(I_{\{h,o,f\}})$. A first set of points X_h is projected onto I_f and I_h via perspective projection π and linked to the corresponding features ϕ_h and ϕ_f to obtain pixel-alignment with the human: $\Phi_{\{h,f\}}^h = \phi_{\{h,f\}}(\pi(X_h, I_{\{h,f\}}))$. Similarly, the pixel-aligned object features $\Phi_{\{o,f\}}^o = \phi_{\{o,f\}}(\pi(X_o, I_{\{o,f\}}))$ are obtained by projecting a different set of points X_o on I_o and I_f and indexing them with the corresponding features ϕ_o and ϕ_f .

3.2.2 Human-object pose priors: In order to learn dense surface details, query points for implicit function need to be sampled within a fixed volume. We leverage a parametric model like SMPL-H for human and a coarse object shape, which can be estimated by any parametric-based human-object reconstruction method during testing, to anchor the 3D spatial location. We compute the neural representations of both the human and the object relative to the SMPL-H center. The object position is defined relative to this center, to learn its 3D spatial relationship with the human. Furthermore, we condition our model on semantic features of the SMPL-H and object template as a form of 3D spatial regularization to encourage plausible reconstructions. To define the human pose features, for a query point $x_h \in X_h$, we look for the closest point x_h^* on SMPL-H, i.e. $x_h^* = \arg \min_{x_h^p} \|x_h - x_h^p\|$ where x_h^p are points on the SMPL-H mesh. Human pose prior features $\sigma_h(x_h) = [d_h, v_h, z_h]$ comprise three elements: a signed distance value d_h between x_h and x_h^* , a visibility label $v_h \in \{1, 0\}$ where v_h indicates if x_h^* is visible in the image when SMPL-H and the object mesh is projected together, and a relative depth-aware feature $z_h = (x_h^z - z_c)$ where z_c is the depth of the SMPL-H center and x_h^z is the depth of the query point x_h . Analogously, for a query point $x_o \in X_o$ and its closest point on the object mesh, x_o^* , the object pose prior features $\sigma_o(x_o) = [d_o, v_o, z_o]$.

3.2.3 Transformer encoders: Image I_f contains spatial cues about human-object relationship, while images $I_{\{h,o\}}$ provide information

about the surface details of the human and object, such as normals and hallucinated occluded regions. To better merge their features, we propose using two attention-based encoders, A_h and A_o . An attention score is computed for two paired images by assessing the compatibility between a query and its corresponding key, producing two feature embeddings:

$$\begin{aligned}\varphi_h &= A_h(\Phi_h^h, \Phi_f^h, \sigma_h(x_h)), \\ \varphi_o &= A_o(\Phi_o^o, \Phi_f^o, \sigma_o(x_o)).\end{aligned}$$

one for the human φ_h and another for the object $\varphi_o = A_o$. The pose prior features $\sigma_{\{h,o\}}$ are also concatenated for conditioning. Thus, the implicit representation of human and object respectively, is estimated by focusing the attention between human-object interaction or respective surface details. Hence, the model is able to reconstruct the human-object such that both their spatial relationship and surface details are accurately captured.

3.2.4 Implicit function estimation: The implicit functions of the human f_h and object f_o are modeled with two separate multi-layer perceptrons (MLPs), which jointly estimate the occupancy values of X_h and X_o :

$$\hat{s} = \begin{cases} \hat{s}_h = f_h(\varphi_h, X_h), & \hat{s}_h \in \mathbb{R}, \\ \hat{s}_o = f_o(\varphi_o, X_o), & \hat{s}_o \in \mathbb{R} \end{cases} \quad (4)$$

where $\hat{s}$ is the implicit representation of the final human-object shape, with $\hat{s}_h$ representing the human component and $\hat{s}_o$ representing the object component.

3.2.5 Training: The neural implicit model is trained end-to-end with the following objective function $\mathcal{L} = \mathcal{L}_h + \mathcal{L}_o$:

$$\mathcal{L}_h = \frac{1}{N_h} \sum_{j=1}^{N_h} \left| f_h(\varphi_h^j, X_h^j) - f_h^{gt}(\varphi_h^j, X_h^j) \right|^2 \quad (5)$$

where N_h is the number of points X_h and f_h^{gt} is the ground-truth implicit function for the human shape, and

$$\mathcal{L}_o = \frac{1}{N_o} \sum_{j=1}^{N_o} \left| f_o(\varphi_o^j, X_o^j) - f_o^{gt}(\varphi_o^j, X_o^j) \right|^2 \quad (6)$$

where N_o is the number of points X_o and f_o^{gt} is the ground-truth implicit function for the object shape.

3.2.6 Inference: During inference, the 3D shape is obtained by extracting iso-surface $f = 0.5$ of the probability field $\hat{s}$ at threshold 0.5 with the Marching Cubes algorithm [Lorensen and Cline 1987].

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets: Due to the lack of 3D human-object datasets with high-quality meshes that represent clothing details, we introduce the dataset synHOR (Fig. 4) to train our model. synHOR is a synthetic dataset created using 526 3D human scans from Thuman 2.0 [Yu et al. 2021] and 27 3D object scans including all 20 objects from BEHAVE [Bhatnagar et al. 2022] and selected 7 from HODome [Zhang et al. 2023a]. We picked random pairs of human and object meshes, where the object was initialized with a random

pose and optimized to be in contact with the human. We then simulated 6 random translations of the human-object pair within the FOV of a perspective camera placed at the origin, and rendered 180 views for each translation. Finally, we discard the images where the object is not in view. We train ReCHOR with 500 subjects of synHOR. For ablation study, we use 99 images from synHOR. We test the generalization power of ReCHOR by reconstructing the human and the object from real images taken from BEHAVE [Bhatnagar et al. 2022]. BEHAVE is a real-world dataset that captures interactions between 8 humans and 20 objects. Due to its real-world setting, high-quality ground truth meshes are unavailable; therefore, we perform only a qualitative evaluation.

4.1.2 Implementation details:

Diffusion model: Inspired by SitH [Ho et al. 2024], our image-conditioned latent diffusion model [Rombach et al. 2022] used to generate missing body regions as described in section 3.1, is trained by fine-tuning the diffusion U-Net’s [Rombach et al. 2022] weights. These weights are initialized using the Zero-1-to-3 [Lugaresi et al. 2019] model, combined with a trainable ControlNet [Zhang et al. 2023b] model, following the default network setups with input channel adjustment. The ControlNet input is a 1 channel mask and the diffusion U-Net input is a 3-channel RGB image of size 512 x 512. We train the model with a batch size of 6 images on a single A100 NVIDIA GPU, with a learning rate of 4×10^{-6} and adopting a constant warmup scheduling. The ControlNet model’s conditioning scale is fixed at 1.0. We use classifier-free guidance in our training, involving a dropout rate of 0.05 for the image-conditioning. During inference, a classifier-free guidance scale of 2.5 is applied to generate the output images.

Attention-based neural implicit model: For each input image of size 2048x1536, we take a crop around the human-object bounding box center and resize it to 512px for the network. We use SAM [Kirillov et al. 2023] to segment the input image. CHORE [Xie et al. 2022] is applied for pose estimation and the 2D normal map is estimated from I_h using pix2pixHD [Wang et al. 2018] as in PIFuHD [Saito et al. 2020] during testing. Following PIFu [Saito et al. 2019], we use a four-stack Hourglass network that yields a 256-dimensional feature map for querying pixel-aligned features.

A standard multi-head transformer-style architecture is applied for the two attention-based encoders A_h and A_o , as shown in Fig. 3. Given three vectors query $Q = M_q\phi$, key $K = M_k\phi$ and value $V = M_v\phi$ as the embedding of the original feature Φ and parameterized by matrices M_q , M_k and M_v , an attention score is computed for each input feature $\Phi_{\{h,f\}}^h$ for A_h and $\Phi_{\{o,f\}}^o$ for A_o based on the compatibility of a query with a corresponding key:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (7)$$

where d_k is the common dimension of K , Q and V . Multiple heads are used to compute features for the human and object:

$$\begin{aligned}\text{MultiHead}(Q, K, V) &= \text{concat}(H_1, \dots, H_h)W^o \\ H_i &= \text{Attention}(QW_i^q, KW_i^k, VW_i^v)\end{aligned} \quad (8)$$

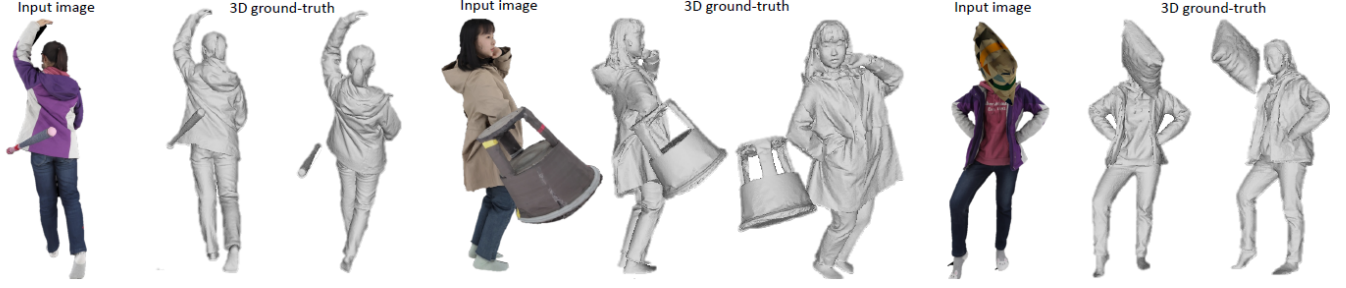


Figure 4: Selected 3D ground-truth human-object meshes from synHOR

where QW_i^q, KW_i^k, VW_i^v are the parameters of Q, K and V , and W^o are the parameters of the final projection. The final feature is computed as the mean of the processed features with each merged feature $\varphi_{\{h,o\}}$ containing local information from the human or object image along with global scene information from the input image.

Finally, to estimate the neural implicit representations, each decoder f_h, f_o is an MLP with the number of neurons (259, 1024, 512, 256, 128, 1) and skip connections at 2nd, 3rd and 4th layers. To train the attention-based neural implicit model, we sample two sets of $N = 200000$ points around the ground-truth object and human surfaces, using a mix of uniform and importance sampling with variances $\sigma = 0.06, 0.01, 0.035$. From these, we randomly select $N_h = N_o = 20000$ points to create the X_h and X_o subsets. These points are projected into the input images following the Kinect camera model from the BEHAVE dataset [Bhatnagar et al. 2022]. We train the model end-to-end using Adam optimizer with a learning rate of $1e-4$ and batch size 4 on a single A100 GPU.

4.1.3 Metrics: For our ablation study, we compute 3D metrics point-to-surface distance (P2S), Chamfer distance (CD), normal consistency (Normal), Intersection over Union (IoU) and fScore [Tatarchenko et al. 2019] on the generated meshes. The metrics are computed between the combined 3D human-object ground truth and the estimated reconstruction, with the estimated shape aligned to the ground truth by rescaling and translating it to match the human centroid.

4.2 Comparisons

4.2.1 Parametric human-object reconstruction methods: We compare ReCHOR against state-of-the-art human-object reconstruction methods from a single image that use a template-based or coarse representation of the 3D shapes, namely CHORE [Xie et al. 2022], CONTHO [Nam et al. 2024] and HDM [Xie et al. 2024a]. As shown in Fig. 5, ReCHOR is the only method that reconstructs realistic clothed human shapes along with the objects. Related works use a parametric-based representation of humans, which lacks detail and realism. These approaches are designed solely to optimize the human-object spatial configuration. ReCHOR leverages them as priors and significantly improves the realism of the reconstruction while preserving the 3D spatial configuration. Note that we do not quantitatively evaluate parametric methods as the quality of their human shapes is much lower than non-parametric shapes.

Table 1: Quantitative comparisons on synHOR dataset between ReCHOR and various baselines.

Methods	synHOR				
	P2S ↓	CD ↓	IoU ↑	Normal ↑	f-Score ↑
2PIFuHD _{ho}	48.29	80.09	.3482	.5347	10.79
PIFuHD _{ho}	43.50	91.97	.3917	.5661	11.00
σ PIFuHD _{ho}	31.66	50.78	.4228	.7090	20.59
2 σ PIFuHD _{ho}	28.82	36.83	.6966	.7196	25.62
2 σ PIFuHD _{ho} ^{sep}	25.83	33.20	.8176	.7204	29.17
2 σ PIFuHD _{ho} ^{all}	23.87	26.53	.8172	.7601	32.72
No Trans	22.90	24.66	.8447	.7683	33.97
Single MLP	17.57	24.70	.8267	.7986	43.98
Single All	15.05	18.12	.8752	.8248	47.71
Sep _{no}	23.36	27.22	.8337	.7523	34.38
Sep	47.69	81.93	.4564	.5868	10.27
ReCHOR (ours)	14.23	17.15	.8902	.8241	48.84

4.3 Ablation Study

We introduce different baselines and alternate architectures, to show the effectiveness of the proposed model explained in section 3.2. Note that for ablation study on synHOR, we use the ground-truth SMPL-H model, whereas for BEHAVE, we use the SMPL-H model as estimated by CHORE [Xie et al. 2022].

4.3.1 Ablation with baseline architecture: We use PIFuHD as baseline to estimate implicit representations for the human and object with an MLP using pixel-aligned features from the input image. We retrain PIFuHD on synHOR (PIFuHD_{ho}) and introduce modifications by varying the input and no. of MLPs. When using a single MLP, a single occupancy is estimated that combines human and object. We then modify the architecture to include two MLPs to estimate an implicit representation of both human and object (2PIFuHD_{ho}). We repeat the experiments by concatenating the pose features σ to the extracted features (indicated with the prefix σ). In another experiment, we give the segmented human and object images (2 σ PIFuHD_{ho}^{sep}) as input and finally, the images (I_f, I_h, I_o) are concatenated together as input for the feature extraction (2 σ PIFuHD_{ho}^{all}).

4.3.2 Ablation with ReCHOR architecture: We train and test 3 alternate configurations of ReCHOR to highlight the importance of using the attention-based neural implicit model.

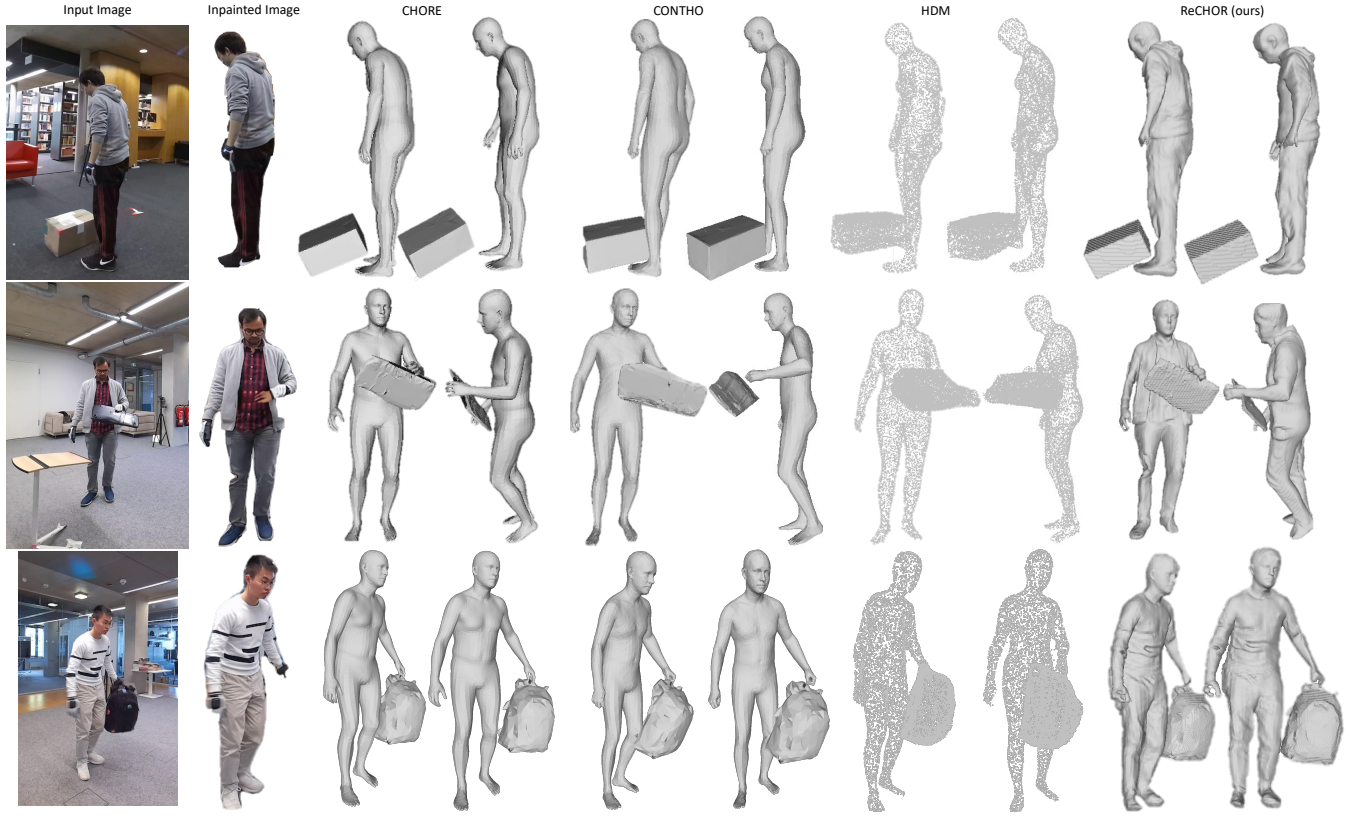


Figure 5: Qualitative evaluations against methods which aim to reconstruct human-object jointly with examples from BEHAVE dataset [Bhatnagar et al. 2022]. Note that HDM generates point clouds rather than meshes. Front and side views are shown.

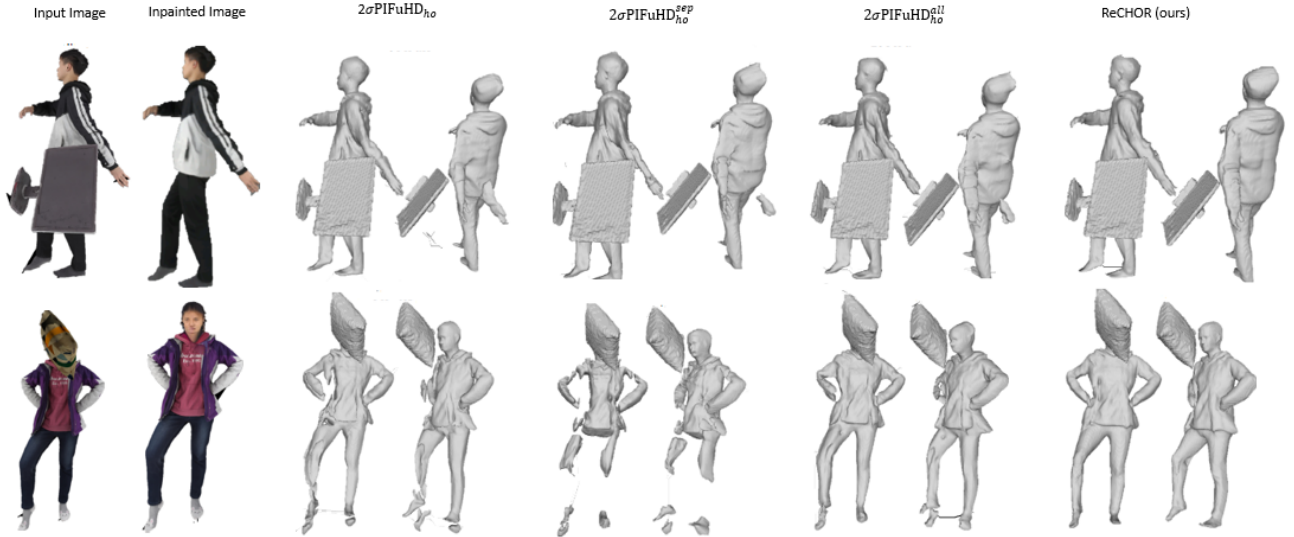


Figure 6: Qualitative results from synHOR showing the effect of using different baseline architectures. Front and side views are shown.

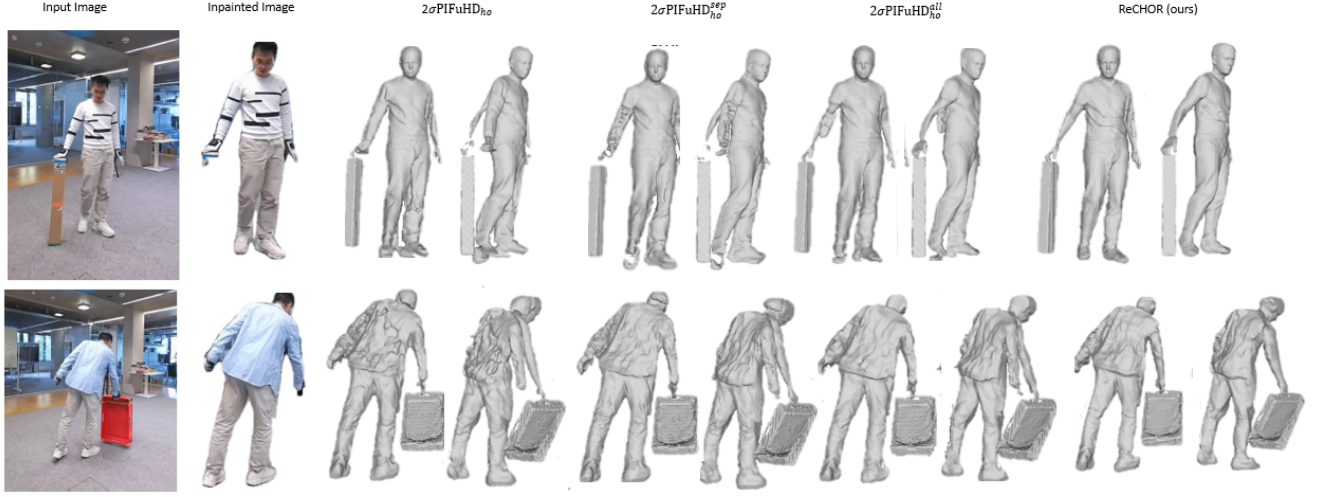


Figure 7: Qualitative results from BEHAVE [Bhatnagar et al. 2022] showing the effect of using different baseline architectures. Front and side views are shown.

- **Single MLP:** A single MLP is used on the concatenated features $\varphi_{\{h,o\}}$ to estimate a single occupancy $\hat{s}$ for both human and object.
- **Single All:** Similar to above, but a single transformer encoder is used.
- **No Trans:** The embeddings $\phi_{\{h,o,f\}}$ are simply concatenated without applying the transformer, $\hat{s}_h$ and $\hat{s}_o$ are estimated from the concatenated features.

4.3.3 Effect of joint reconstruction: An alternative to reconstructing human and object jointly, is to reconstruct them separately with no knowledge of each other. In other words, to reconstruct the human for example, the object will first be cropped out, and then the occluded regions need to be inpainted. However, such an approach has many disadvantages as it cannot model the spatial relationships, with no context of the other entity, and it is highly dependent on the accuracy of inpainting. We illustrate this using the following approach.

- **Sep_{no}:** Two separate networks are trained. One reconstructs the human shape using the full-body diffusion model generated human image and the human pose feature. Similarly, the other reconstructs the object shape. No transformer encoder is used.
- **Sep:** Two separate networks trained, similar to Sep_{no}, but the input RGB image is also considered. Each network extracts features from the input RGB image and merges them with either the full-body human or object features using transformer encoders.

Both quantitative (Tab. 1) and various qualitative (Fig. 6, Fig. 7, Fig. 8, Fig. 9) results show that ReCHOR has the best architecture, demonstrating its design philosophy. It significantly outperforms the baseline ($2\sigma\text{PIFuHD}_{ho}^{all}$). The difference among the various baseline architectures indicate that the performance of the model improves when additional inputs, like pose priors and segmented human, object views, are provided. However, ReCHOR demonstrates

Table 2: Quantitative results on synHOR dataset obtained with different combinations of input data.

Methods	synHOR				
	P2S ↓	CD ↓	IoU ↑	Normal ↑	f-Score ↑
Only RGB	47.86	78.25	.4472	.5962	11.33
No $\sigma_{h,o}$	44.14	71.72	.5064	.6175	12.85
No S_N	14.24	17.12	.8661	.8024	41.46
ReCHOR (ours)	14.23	17.15	.8902	.8241	48.84

how these inputs are combined matters. When no transformer is applied (**No Trans**), it performs only marginally better than the baseline, ($2\sigma\text{PIFuHD}_{ho}^{all}$). When a single transformer is applied (**Single All**, **Single MLP**), due to common decoder, the interaction between the human and object is not well modeled, as seen from the hand position of the qualitative BEHAVE image result. Finally, training the human and object separately, (**Sep_{no}**, **Sep**), perform significantly worse than ReCHOR, as they cannot model human-object spatial relationship.

4.3.4 Effect of different inputs: In this ablation study, we study the impact of those inputs to ReCHOR, which do not require a change in its architecture.

- **No S_N :** Same as ReCHOR but without 2D normal map S_N concatenated to I_h .
- **No $\sigma_{\{h,o\}}$:** Same as ReCHOR but without processing the human-object pose features $\sigma_{\{h,o\}}$.
- **Only RGB:** Same as ReCHOR without 2D normal map S_N and human-object pose features $\sigma_{\{h,o\}}$.

Fig. 10 shows examples of human-object shapes reconstructed using different combinations of input, with quantitative results in Tab. 2. When normal maps are excluded (No S_N), smoother shapes are reconstructed, with slight drop in the quality of the reconstruction. Omitting pose features (No $\sigma_{h,o}$), have a major impact on the reconstruction, and introduces depth ambiguity. This is especially

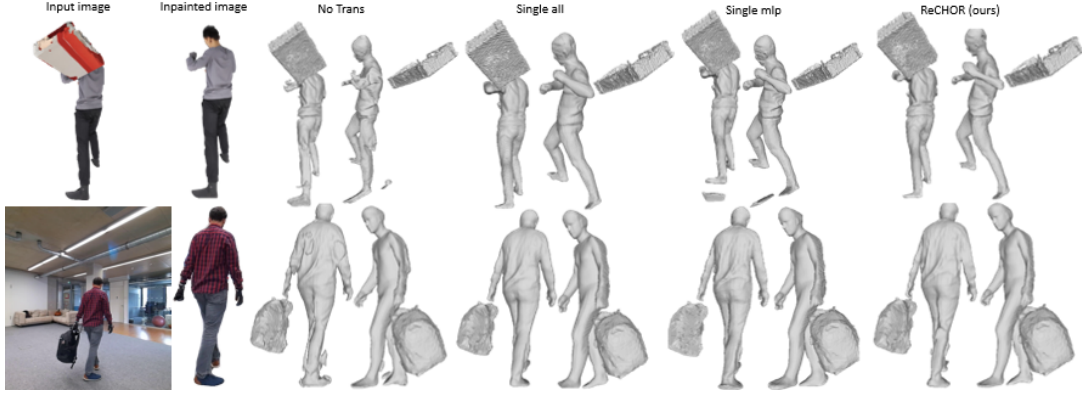


Figure 8: Qualitative results showing the effect of using different configurations of ReCHOR. Front and side views are shown from synHOR (top) and BEHAVE [Bhatnagar et al. 2022] (bottom).

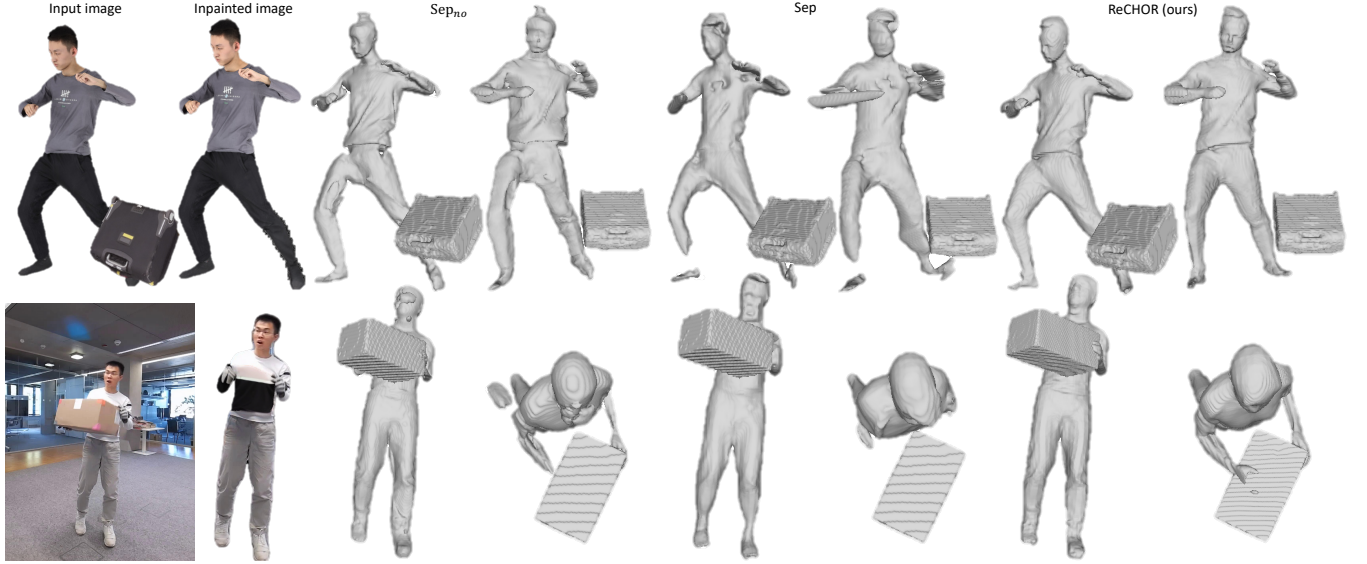


Figure 9: Qualitative results obtained when the human and object is reconstructed in isolation. Front and side views are shown from synHOR in the top row, from BEHAVE [Bhatnagar et al. 2022] in the bottom.

true for objects, since all objects do not have a common template shape, unlike humans. (Only RGB) has the worst reconstruction due to absence of both pose and normals.

4.3.5 Effect of inpainting: In scenes containing both humans and objects, the object often occlude parts of the human body. When only partial regions of the body are visible, the network fails to reconstruct the occluded regions. If the partial human image is directly input into the attention-based neural implicit model, it will fail to reconstruct these occluded regions, producing holes in human shapes. Inpainting the missing body regions in the input image is crucial to reconstruct realistic clothed human and object shapes.

ReCHOR addresses the challenge of human occlusion by leveraging the generative capability of diffusion models to inpaint the occluded body regions. A fine-tuning strategy is adopted to optimize

Table 3: Quantitative evaluation of changing the diffusion module of ReCHOR. The human-object shapes reconstructed with the attention-based neural implicit module of ReCHOR are used for evaluation.

Methods	synHOR				
	P2S ↓	CD ↓	IoU ↑	Normal ↑	f-Score ↑
No inpainting	19.19	22.24	.8679	.8080	40.77
MAT	15.39	18.94	.8841	.8132	46.44
Normal	15.18	17.70	.8860	.8163	47.02
Both	14.91	18.49	.8859	.8160	48.26
ReCHOR (ours)	14.23	17.15	.8902	.8241	48.84

the cross-attention layers of a pre-trained diffusion model, conditioning it on the mask of the body regions requiring inpainting. In

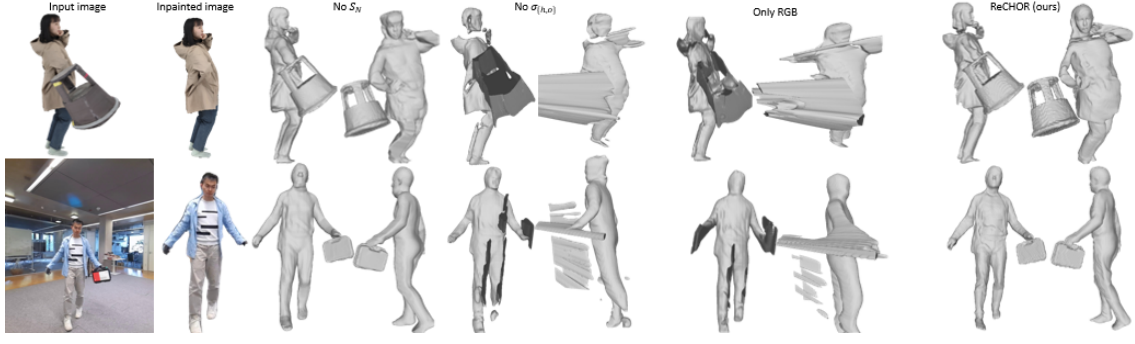


Figure 10: Qualitative results showing the effect of using different combinations of input data. Front and side views are shown from synHOR in the top row, and on BEHAVE [Bhatnagar et al. 2022] in the bottom row.



Figure 11: Human-object shapes reconstructed by changing the diffusion module of ReCHOR. Front view with and without the object and side views without the object are shown from synHOR in the top row, from BEHAVE [Bhatnagar et al. 2022] in the bottom.

this section, we explore different solutions for inpainting, including MAT [Li et al. 2022], a transformer-based model for large hole inpainting, conditioning the diffusion model with the SMPL [Romero et al. 2017] normal map (Normal), and conditioning with both the mask and the normal map (Both).

The effect of using these approaches for the inpainting module of ReCHOR is quantitatively evaluated by estimating the implicit representation of human-object shapes using their generated full-body human image (Tab. 3). Examples of human-object reconstructions using these methods for inpainting are shown in Fig. 11. As expected, the worst results occur when no inpainting is applied (No inpainting). MAT struggles to produce realistic body regions in the RGB images, propagating noise into the final reconstruction. Conditioning the diffusion model with both the SMPL normal map and the mask also degrades performance. The best results are achieved when either the SMPL normal or the mask of the occluded regions is processed using ControlNet. In particular, processing only the mask proves more robust, as seen in the reconstructed human-object shape in the top row of Fig. 11, where unnatural black regions in the full-body human image generated with Normal configuration caused gaps in the reconstructed shape.

Note that we do not quantitatively evaluate the generated images against ground truth, since the main goal of ReCHOR is 3D reconstruction rather than inpainting.

5 Limitations and Future Work

Modeling interactions between realistic clothed humans and objects is an extremely challenging problem. This work represents the first step towards achieving this goal and thus has certain limitations. Since ReCHOR relies on parametric pose priors, inaccuracies in these priors can propagate to the final reconstruction, compromising the accuracy of human-object interactions. In the future, we would like to improve the accuracy. To achieve that goal, we would further develop a clothed human-object dataset that represents more plausible real-world interactions than synHOR. We also aim to improve the quality of the object shape by introducing higher quality ground-truth object shapes in the dataset to learn high-frequency details for the objects as well. While our approach specifically addresses cases where a human is occluded by an object, we will extend it to handle object occlusions as well. We also aim to retrieve textures for clothed human-object reconstructions.

6 Conclusion

We presented ReCHOR, the first novel method that addresses the complex task of jointly reconstructing realistic clothed humans and objects from a human-object scene image. ReCHOR combines the power of attention-based neural implicit learning with a generative diffusion model and pose priors. Our experiments show that our method generalizes well to a real world dataset on which it was not trained and outperforms the state-of-the-art methods in reconstructing realistic human-object scenes. This is particularly valuable for media-related applications, where the goal is to achieve realism rather than rely on coarse shapes.

References

- Mohammad Samiul Arshad and William J Beksi. 2023. LIST: Learning Implicitly from Spatial Transformers for Single-View 3D Reconstruction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 9321–9330.
- Bharat Lal Bhatnagar, Xianghui Xie, Ilya A Petrov, Cristian Sminchisescu, Christian Theobalt, and Gerard Pons-Moll. 2022. Behave: Dataset and method for tracking human object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 15935–15946.
- Rongshan Chen, Yuancheng Yang, and Chao Tong. 2023. G2Fu: Graph-based implicit function for single-view 3D reconstruction. *Engineering Applications of Artificial Intelligence* 124 (2023), 106493.
- Yixin Chen, Siyuan Huang, Tao Yuan, Siyuan Qi, Yixin Zhu, and Song-Chun Zhu. 2019. Holistic++ scene understanding: Single-view 3d holistic scene parsing and human pose estimation with human-object interaction and physical commonsense. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 8648–8657.
- Zhiqin Chen and Hao Zhang. 2019. Learning implicit fields for generative shape modeling. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 5939–5948.
- George Fahim, Khalid Amin, and Sameh Zarif. 2022. Enhancing single-view 3D mesh reconstruction with the aid of implicit surface learning. *Image and Vision Computing* 119 (2022), 104377.
- Tong He, John Collomosse, Hailin Jin, and Stefano Soatto. 2020. Geo-pifu: Geometry and pixel aligned implicit functions for single-view human reconstruction. *Advances in Neural Information Processing Systems* 33 (2020), 9276–9287.
- I Ho, Jie Song, Otmar Hilliges, et al. 2024. Sith: Single-view textured human reconstruction with image-conditioned diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 538–549.
- Yinghao Huang, Omid Taheri, Michael J. Black, and Dimitrios Tzionas. 2024. InterCap: Joint Markerless 3D Tracking of Humans and Objects in Interaction from Multi-view RGB-D Images. *International Journal of Computer Vision (IJCV)* (2024). doi:10.1007/s11263-024-01984-1
- Yuheng Jiang, Suyi Jiang, Guoxing Sun, Zhuo Su, Kaiwen Guo, Minye Wu, Jingyi Yu, and Lan Xu. 2022. Neuralhofusion: Neural volumetric rendering under human-object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 6155–6165.
- Yuheng Jiang, Kaixin Yao, Zhuo Su, Zhehao Shen, Haimin Luo, and Lan Xu. 2023. Instant-NVR: Instant Neural Volumetric Rendering for Human-object Interactions from Monocular RGBD Stream. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 595–605.
- Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. 2023. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 4015–4026.
- Nikos Kolotouros, Thimo Alldieck, Enric Corona, Eduard Gabriel Bazavan, and Cristian Sminchisescu. 2024. Instant 3D Human Avatar Generation using Image Diffusion Models. *arXiv preprint arXiv:2406.07516* (2024).
- Nupur Kumari, Bingliang Zhang, Richard Zhang, Eli Shechtman, and Jun-Yan Zhu. 2023. Multi-concept customization of text-to-image diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 1931–1941.
- Manyi Li and Hao Zhang. 2021. D2im-net: Learning detail disentangled implicit fields from single images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 10246–10255.
- Wenbo Li, Zhe Lin, Kun Zhou, Lu Qi, Yi Wang, and Jiaya Jia. 2022. MAT: Mask-Aware Transformer for Large Hole Image Inpainting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Shichen Liu, Shunsuke Saito, Weikai Chen, and Hao Li. 2019. Learning to infer implicit surfaces without 3d supervision. *Advances in Neural Information Processing Systems* 32 (2019).
- William E Lorensen and Harvey E Cline. 1987. Marching cubes: A high resolution 3D surface construction algorithm. *ACM siggraph computer graphics* 21, 4 (1987), 163–169.
- Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Yong, Juhyun Lee, et al. 2019. Mediapipe: A framework for perceiving and processing reality. In *Third workshop on computer vision for AR/VR at IEEE computer vision and pattern recognition (CVPR)*, Vol. 2019.
- Lars Mescheder, Michael Oechsle, Michael Niemeyer, Sebastian Nowozin, and Andreas Geiger. 2019. Occupancy networks: Learning 3d reconstruction in function space. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4460–4470.
- Hyeonjin Nam, Daniel Sungho Jung, Gyeongsik Moon, and Kyoung Mu Lee. 2024. Joint Reconstruction of 3D Human and Object via Contact-Based Refinement Transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Jeong Joon Park, Peter Florence, Julian Straub, Richard Newcombe, and Steven Lovegrove. 2019. DeepSDF: Learning continuous signed distance functions for shape representation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 165–174.
- Marco Pesavento, Marco Volino, , and Adrian Hilton. 2022. Super-resolution 3D Human Shape from a Single Low-Resolution Image. In *ECCV*.
- Marco Pesavento, Marco Volino, and Adrian Hilton. 2025. COSMU: Complete 3D human shape from monocular unconstrained images. In *European Conference on Computer Vision*. Springer, 201–219.
- Marco Pesavento, Yuanlu Xu, Nikolaos Sarafianos, Robert Maier, Ziyang Wang, Chun-Han Yao, Marco Volino, Edmond Boyer, Adrian Hilton, and Tony Tung. 2024. ANIM: Accurate Neural Implicit Model for Human Reconstruction from a single RGB-D image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5448–5458.
- Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. 2022. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125* 1, 2 (2022), 3.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10684–10695.
- Javier Romero, Dimitrios Tzionas, and Michael J. Black. 2017. Embodied Hands: Modeling and Capturing Hands and Bodies Together. *ACM Transactions on Graphics, (Proc. SIGGRAPH Asia)* 36, 6 (Nov. 2017).
- Shunsuke Saito, Zeng Huang, Ryota Natsume, Shigeo Morishima, Angjoo Kanazawa, and Hao Li. 2019. Pifu: Pixel-aligned implicit function for high-resolution clothed human digitization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2304–2314.
- Shunsuke Saito, Tomas Simon, Jason Saragih, and Hanbyul Joo. 2020. Pifuhd: Multi-level pixel-aligned implicit function for high-resolution 3d human digitization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 84–93.
- Akash Sengupta, Thimo Alldieck, Nikos Kolotouros, Enric Corona, Andrei Zanfir, and Cristian Sminchisescu. 2024. DiffHuman: Probabilistic Photorealistic 3D Reconstruction of Humans. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 1439–1449.
- Guoxing Sun, Xin Chen, Yizhang Chen, Anqi Pang, Pei Lin, Yuheng Jiang, Lan Xu, Jingyi Yu, and Jingya Wang. 2021. Neural free-viewpoint performance rendering under complex human-object interactions. In *Proceedings of the 29th ACM International Conference on Multimedia*. 4651–4660.
- Maxim Tatarchenko, Stephan R Richter, René Ranftl, Zhuwen Li, Vladlen Koltun, and Thomas Brox. 2019. What do single-view 3d reconstruction networks learn?. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 3405–3414.
- Jianguan Wang, Huanqiang Xu, Xinrui Hu, and Biao Leng. 2023. IFKD: Implicit field knowledge distillation for single view reconstruction. *Mathematical Biosciences and Engineering* 20, 8 (2023), 13864–13880.
- Ting-Chun Wang, Ming-Yu Liu, Jun-Yan Zhu, Andrew Tao, Jan Kautz, and Bryan Catanzaro. 2018. High-resolution image synthesis and semantic manipulation with conditional gans. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 8798–8807.
- Xi Wang, Gen Li, Yen-Ling Kuo, Muhammed Kocabas, Emre Aksan, and Otmar Hilliges. 2022. Reconstructing action-conditioned human-object interactions using commonsense knowledge priors. In *2022 International Conference on 3D Vision (3DV)*. IEEE, 353–362.
- Xianghui Xie, Bharat Lal Bhatnagar, Jan Eric Lenssen, and Gerard Pons-Moll. 2024a. Template Free Reconstruction of Human-object Interaction with Procedural Interaction Generation. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Xianghui Xie, Bharat Lal Bhatnagar, and Gerard Pons-Moll. 2022. Chore: Contact, human and object reconstruction from a single rgb image. In *European Conference on Computer Vision*. Springer, 125–145.

- Xianghui Xie, Bharat Lal Bhatnagar, and Gerard Pons-Moll. 2023. Visibility Aware Human-Object Interaction Tracking from Single RGB Camera. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Xianghui Xie, Jan Eric Lenssen, and Gerard Pons-Moll. 2024b. InterTrack: Tracking Human Object Interaction without Object Templates. *arXiv preprint arXiv:2408.13953* (2024).
- Yuliang Xiu, Jinlong Yang, Xu Cao, Dimitrios Tzionas, and Michael J Black. 2023. ECON: Explicit Clothed humans Optimized via Normal integration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 512–523.
- Yuliang Xiu, Jinlong Yang, Dimitrios Tzionas, and Michael J Black. 2022. Icon: Implicit clothed humans obtained from normals. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 13286–13296.
- Qiangeng Xu, Weiyue Wang, Duygu Ceylan, Radomir Mech, and Ulrich Neumann. 2019. Disn: Deep implicit surface network for high-quality single-view 3d reconstruction. *Advances in neural information processing systems* 32 (2019).
- Xiang Xu, Hanbyul Joo, Greg Mori, and Manolis Savva. 2021. D3d-hoi: Dynamic 3d human-object interactions from videos. *arXiv preprint arXiv:2108.08420* (2021).
- Tao Yu, Zerong Zheng, Kaiwen Guo, Pengpeng Liu, Qionghai Dai, and Yebin Liu. 2021. Function4d: Real-time human volumetric capture from very sparse consumer rgbd sensors. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 5746–5756.
- Zehao Yu, Songyou Peng, Michael Niemeyer, Torsten Sattler, and Andreas Geiger. 2022. Monosdf: Exploring monocular geometric cues for neural implicit surface reconstruction. *Advances in neural information processing systems* 35 (2022), 25018–25032.
- Juze Zhang, Haimin Luo, Hongdi Yang, Xinru Xu, Qianyang Wu, Ye Shi, Jingyi Yu, Lan Xu, and Jingya Wang. 2023a. NeuralDome: A neural modeling pipeline on multi-view human-object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8834–8845.
- Jason Y Zhang, Sam PePOSE, Hanbyul Joo, Deva Ramanan, Jitendra Malik, and Angjoo Kanazawa. 2020. Perceiving 3d human-object spatial arrangements from a single image in the wild. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XII* 16. Springer, 34–51.
- Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. 2023b. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 3836–3847.
- Yanqiang Zhang, Lijin Fang, and Chengpeng Li. 2021. Generalized Deep Implicit Surface Network for Image-based Three-dimensional Object Reconstruction. In *2021 China Automation Congress (CAC)*. IEEE, 276–281.
- Zerong Zheng, Tao Yu, Yebin Liu, and Qionghai Dai. 2021. Pamir: Parametric model-conditioned implicit representation for image-based human reconstruction. *IEEE transactions on pattern analysis and machine intelligence* 44, 6 (2021), 3170–3184.